Gridlock 2.0

Spatio-Temporal Traffic-Demand Forecasting

Technical Solution Report

Best evaluation score

91.25092

Final model: BC blend = 50/50 average of (B) feature-enhanced LightGBM and (C) daytime-proxy-gated LightGBM + XGBoost blend

Key innovation: a daytime validation proxy + disciplined model diversity

HackerEarth ML Hackathon -- Gridlock 2.0

1. Executive Summary

Gridlock 2.0 is a spatio-temporal traffic-demand forecasting problem: predict a bounded congestion 'demand' value in (0, 1] for every (geohash cell, time-of-day) on the daytime portion of Day 49, given the full history of Day 48 and the early-morning of Day 49. Our final solution is the BC blend -- a 50/50 average of a feature-enhanced LightGBM (B) and a model-diversity, daytime-proxy-gated LightGBM+XGBoost blend (C) -- which reached an evaluation score of 91.25092, our best result.

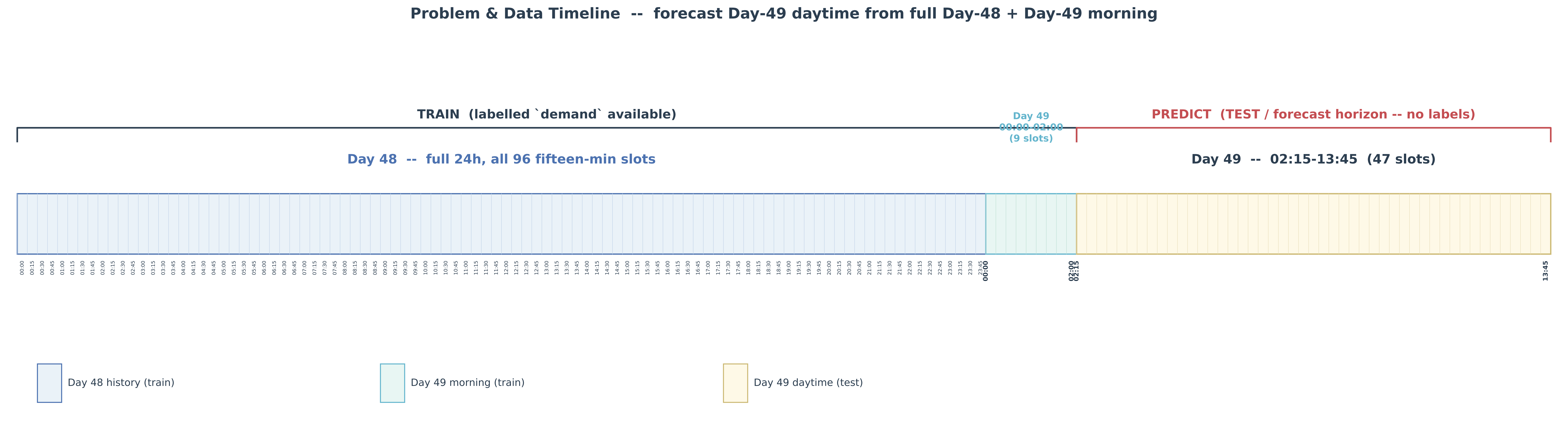
The key innovation was not a single exotic model but a measurement fix plus disciplined model diversity. Early iterations regressed because the only offline cross-validation available was night-dominated and did not track the public evaluation score. We introduced a DAYTIME VALIDATION PROXY -- the first trustworthy offline daytime metric -- and gated every subsequent decision on it. Combined with bias-variance-justified blending of two diverse models, this moved the score from 90.791 (v1) to 91.25092 (BC) in a fully reproducible, leakage-free pipeline.

2. Problem Understanding

Task: supervised regression of a continuous, strictly positive, bounded target demand in (0, 1] -- heavily right-skewed. Metric: RMSE, surfaced as an evaluation score of the form 100 x (1 - RMSE); minimising RMSE is therefore equivalent to maximising the displayed score, so we optimise RMSE directly and clip predictions to [0, 1].

Business objective: forecast short-horizon traffic demand per spatial cell (geohash) and time-of-day so that congestion ('gridlock') can be anticipated and mitigated and resources allocated ahead of the peak.

Forecast structure (the crux of the design): the training data contains Day 48 in full (all 96 fifteen-minute slots) and Day 49 only for the early morning (00:00-02:00). The test set is Day 49 daytime (02:15-13:45). We are therefore predicting the rest of Day 49 from the previous full day plus the current morning -- a genuine temporal forecast, which dictates both the feature design (a day-over-day lag) and the validation design (a daytime proxy).


Problem & Data Timeline -- forecast Day-49 daytime from full Day-48 + Day-49 morning
TRAIN (labelled `demand` available)
Day 49
00:00-02:00
(9 slots)
PREDICT (TEST / forecast horizon -- no labels)
Day 48 -- full 24h, all 96 fifteen-min slots
Day 49 -- 02:15-13:45 (47 slots)
00:00
02:00
02:15
13:45
Day 48 history (train)
Day 49 morning (train)
Day 49 daytime (test)

3. Exploratory Data Analysis

Shapes: train 77,299 x 11, test 41,778 x 10. The target is right-skewed (skew ~3.73, median 0.048, mean 0.094) with no zeros, negatives, or NaN -- so the target is best left raw and clipped, rather than log/Tweedie transformed.

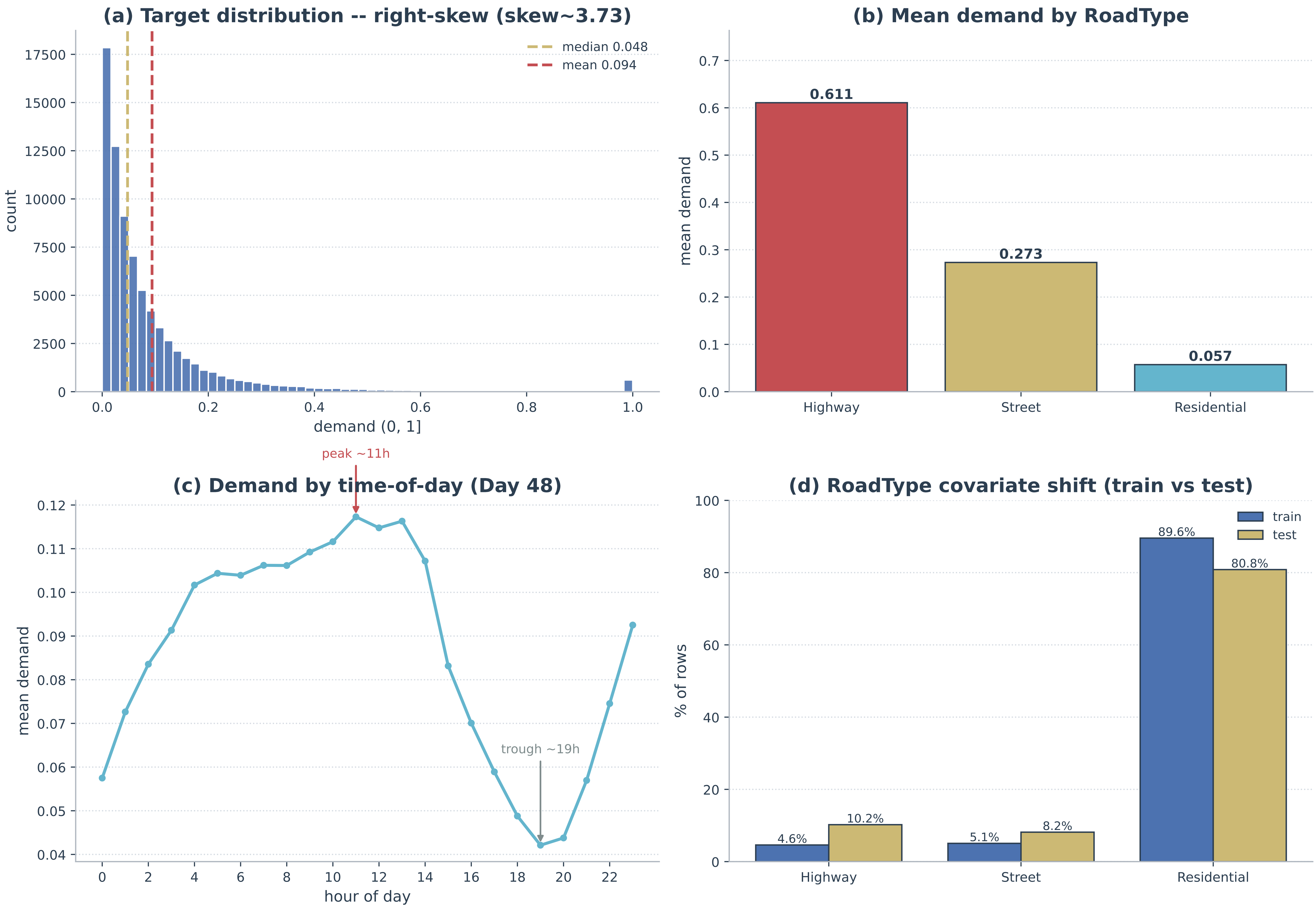
Temporal split: Day 48 has all 96 slots; Day 49 in train is morning-only; the Day-49 daytime window is the test horizon. Granularity is 15 minutes. Geohashes are 6-character cells (~1,249 in train); roughly 10-15 test cells are cold-start (unseen in train), but all of them share a 4- and 5-character prefix present in train, enabling spatial back-off.

Missingness is mild and identical across train and test -- Temperature 3.23%, Weather 1.03%, RoadType 0.78% -- i.e. MCAR-like, so simple imputation plus missingness flags is safe. There are NO duplicates (full-row, feature, or key). The data-quality audit PASSED: Temperature ranges a plausible -14.9..48.3 C, lanes 1-5, geohash length uniform. The dataset is clean -- a finding that, as it turned out, argued AGAINST aggressive balancing or target reshaping.

Covariate shift: the test set is enriched in Highway (4.6% -> 10.2%) and Street (5.1% -> 8.2%) at the expense of Residential (89.6% -> 80.8%). Because Highway demand (~0.61) far exceeds Residential (~0.057), the test mean is structurally higher; the model must lean on RoadType and NumberofLanes, both of which ARE provided in the test set.

Strong predictors: RoadType and NumberofLanes (4-5 lanes ~ highways ~0.60 demand) and a clear time-of-day cycle (peak ~11-13h, trough ~19h). LargeVehicles is essentially fully determined by NumberofLanes (collinear). Weather and Temperature are near-useless (correlation ~0). These facts shaped a tree-based model with cyclical-time and history features.

Exploratory Data Analysis -- key findings



4. Data Processing & Feature Engineering

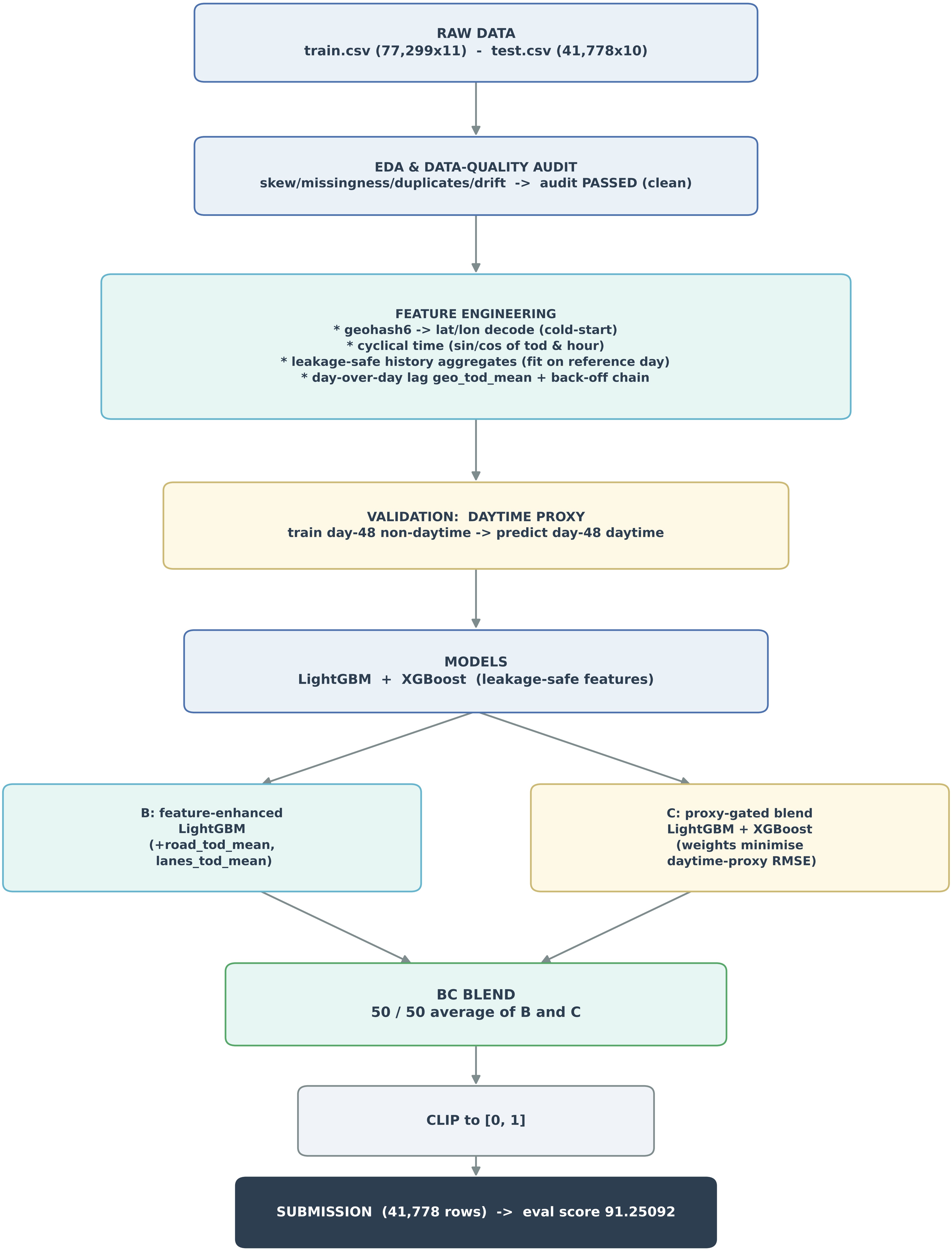
Spatial: each 6-char geohash is decoded to a (lat, lon) cell centre, giving the model a continuous spatial coordinate that generalises to cold-start cells (rather than treating geohash purely as an opaque category).

Temporal: time-of-day in minutes since midnight, hour and minute, and cyclical sin/cos encodings of both tod (period 1,440 min) and hour (period 24h) so the model sees midnight and 23:45 as adjacent.

Leakage-safe history aggregates: per-geohash demand mean/std/median/min/max/count, a globally-smoothed per-geohash mean (geo_mean_smooth), per-time-of-day mean/std, per-RoadType mean, and a geohash frequency. These are TARGET-derived and therefore leakage-prone, so they are fit on a reference day only (Day 48) and applied to the target frame -- never fit on the rows being predicted.

Approach B adds two interaction profiles on top of this base: road_tod_mean (RoadType x time-of-day) and lanes_tod_mean (NumberofLanes x time-of-day), each a smoothed target mean fit leakage-free on the reference rows. These were the only added features that earned their place on the daytime proxy (+0.00388 RMSE on Proxy-T); region, ratio and geo_hour families were ablated and dropped.

End-to-End Pipeline / Architecture



5. How We Filled the Lags (judges asked)

The single strongest signal is the day-over-day lag geo_tod_mean: Day-48 demand at the SAME (geohash, time-of-day). For a Day-49 row this is literally 'what happened here at this time yesterday'. It is present for ~88.9% of test cells. Crucially, it is leakage-free: the aggregate is fit on the prior day (Day 48) and the lag is populated only cross-day (for Day-49 rows), never within the same day being predicted.

When the exact (geohash, time-of-day) lag is missing -- cold-start cells, or a slot never observed on the reference day -- we do NOT drop to a global constant. Instead a graceful BACK-OFF CHAIN fills it, from most-specific to most-general:

- geo_tod_mean (Day-48 demand at the exact geohash + time-of-day)
- -> geo_mean_smooth (that geohash's smoothed mean demand, regularised toward the global mean)
- -> tod_mean (the demand profile for that time-of-day across all cells)
- -> global mean (final fallback)

This degrades gracefully: a brand-new cell still gets a sensible time-of-day-aware estimate, and a never-seen slot still gets the cell's own level. The chain introduces zero leakage because every component is fit on the reference/prior day only.

6. Validation Methodology (the key insight)

This is where the competition was won or lost. The training set has NO daytime labels for Day 49 (the Day-49 train rows are night-only, 00:00-02:00), while the SCORED test is Day-49 daytime. Our initial offline cross-validation -- CV-A (validate on Day-49 night) and a geohash GroupKFold CV-B -- therefore measured the wrong regime. They did not track the public evaluation score and actively misled early iterations (v3, v4 below).

The fix is a DAYTIME PROXY. Day 48 is the only day with daytime labels, so we train on Day-48 NON-daytime rows and predict Day-48 DAYTIME rows -- the same time-of-day window as the test. This is the first offline metric that actually exercises the scored horizon. It is pessimistic in absolute RMSE (Day 48 has no prior day, so the day-over-day lag is absent inside the proxy), but every candidate carries the identical handicap, making it a trustworthy RELATIVE comparator. Final boosting-round counts were calibrated separately on the lag-present CV-A protocol that produced v1. From this point on, every modelling decision was gated on the daytime proxy.

Validation Strategy -- why night CV misled us, and the daytime-proxy fix

MISLEADING: Night-only CV (CV-A / geohash CV-B)

validates on NIGHT labels -- a different regime from the scored DAYTIME test

Day 48 (all slots)	D49 night 00:00-02:00	D49 daytime 02:15-13:45
TRAIN	VALIDATE	TEST (scored)

X Did NOT track the evaluation score

Night CV ranked v3 morning/shift features and v4's 10-seed bag as 'wins' -- yet neither improved the daytime evaluation score (90.678 / 90.689 < 90.791). The offline objective never saw the horizon being scored.

TRUSTWORTHY: Daytime proxy

validate on the ONLY available daytime labels (Day-48 daytime slots)

Day 48 non-daytime (train fold)	Day 48 daytime (predict)	D49 daytime test
TRAIN	VALIDATE (daytime!)	MIRRORS this

OK First trustworthy daytime metric

Train on Day-48 non-daytime rows, predict Day-48 daytime rows -- same time-of-day window as the test. Pessimistic in absolute RMSE (no day-over-day lag) but a reliable RELATIVE comparator. Every modelling decision (B, C, BC) was gated on this proxy.

7. Model Evolution & Final Model Selection

We record each explored candidate, because the comparisons directly shaped the final design and show how the solution evolved (numbers are reported honestly):

- v1 -- single LightGBM on leakage-safe features (217 rounds): evaluation score 90.791. A solid, honest baseline; the day-over-day lag does most of the work.
- v3 -- added morning-anchor / day_shift / scaled_lag features + a blend: evaluation score 90.678. An explored candidate that exposed a night-CV artefact -- features built to extrapolate night->day flattered the night CV but did not transfer to the daytime test. Lesson: night CV misleads.
- v4 -- 10-seed bag at fewer rounds (a sub-seed override cut effective rounds): evaluation score 90.689. An explored candidate that underfit. Lesson: match the round count; don't override LightGBM's internal sub-seeds.
- v5 -- seed-averaging at matched rounds: ~90.689, i.e. within noise of v1. Confirmed v1 was already near the ceiling of that single-model recipe.

Breakthrough: we introduced the daytime proxy and then ran five parallel approaches gated on it -- A (HPO: found v1 already near-optimal), B (road_tod / lanes_tod features: a real daytime-proxy gain), C (model diversity, LightGBM + XGBoost, proxy-gated blend), D (residual/ratio target reformulation: parity, no gain), and E (B+C stack: did not compound).

- C -> evaluation score 91.186: the first real break. Adding XGBoost's diversity moved the high-demand highway predictions that dominate the RMSE -- precisely the rows a single LightGBM was systematically missing.
- BC = 50/50 average of B and C -> evaluation score 91.25092 (the selected final model). B and C are diverse but highly correlated (corr ~0.998), so a simple average shaves variance on the high-demand cells without changing the bias -- a textbook variance-reduction win.


Model Evolution & Final Model Selection
Evaluation Score (100 × (1 - RMSE))
91.4
91.2
91.0
90.8
90.6
90.4
Selected Final Model
91.25092
91.186
model diversity
breakthrough
v1 baseline 90.791
90.791
90.678
90.689
90.689
explored candidates
(night-CV gated)
v1
(LGBM
baseline)
v3
(morning/
shift)
v4
(10-seed
bag)
v5
(seed-avg)
C
(LGBM+XGB
blend)
BC
(50/50
B+C)

8. Why BC Is Best & Reproducible

BC is the best submission for principled, not lucky, reasons. It is leakage-free (all target-derived aggregates fit on a reference day only; the lag populated cross-day only), daytime-validated (every component was gated on the daytime proxy, not the misleading night CV), and deterministic (fixed seeds, deterministic splits, seed-averaged models). The blend itself is justified by bias-variance reasoning -- averaging two diverse-but-correlated estimators reduces variance on the heavy-tailed high-demand cells -- rather than by chasing the public evaluation score. The reproduce.py package in the parent final_submission folder regenerates the submission end-to-end.

9. Results

Model evolution and what moved the needle:

```
version   eval score   note
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v1        90.79100     LightGBM baseline (leakage-safe features)
v3        90.67800     morning/shift feats  -> explored candidate
v4        90.68900     10-seed bag, few rounds -> explored candidate
v5        90.68900     seed-avg, matched rounds (~v1)
C         91.18600     LGBM+XGB proxy-gated blend (diversity)
BC        91.25092     50/50 average of B and C  <- selected final model
```

What HELPED: model diversity (XGBoost alongside LightGBM) and prediction blending (the BC average). What did NOT help: extra features beyond road_tod/lanes_tod, hyper-parameter optimisation (v1 was already near-optimal), target reformulation (residual/ratio), and class/target balancing. The recurring lesson: the data was clean and the target was best left raw -- gains came from honest validation and diversity, not from reshaping the problem.

10. Conclusion & Reproducibility

The Gridlock 2.0 solution shows that on a clean, well-structured forecasting problem the decisive levers are (1) a validation signal that matches what is actually scored, and (2) model diversity blended with bias-variance discipline. The daytime proxy converted a blind, score-chasing process into a measured one; model diversity plus a 50/50 blend then delivered the 91.25092 best score.

Reproducibility statement: the pipeline uses fixed global seeds and deterministic splits; all target-derived encoders/aggregates are fit on reference (prior-day) rows only; predictions are clipped to [0, 1]; and dependency versions are pinned. The parent final_submission/reproduce.py regenerates the BC submission deterministically from the raw train/test CSVs.